PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: LogiMind - Logistics Software

Version: 2.0

1. Project Overview

- **Problem Statement:** SME logistics rely on manual spreadsheets, static routing, and reactive communication. Lack of real-time visibility into weather disruptions, fuel volatility, and traffic delays causes missed time windows, inflated costs, and poor customer trust. Enterprise-grade TMS platforms with predictive analytics are financially out of reach for small operators.
- **Target Domain:** AI-native logistics decision support and dynamic route optimization for Small & Medium Enterprises (SMEs).
- **Proposed Solution Summary:** LogiMind is a web-based logistics co-pilot that separates route calculation from contextual reasoning. A deterministic routing engine generates baseline paths, while an LLM agent evaluates them against live external data (weather, fuel prices, news) and business constraints. The system outputs optimized delivery sequences, risk alerts, cost impact analysis, and ready-to-send customer notifications in the dashboard display, shifting SME operations from reactive guesswork to proactive execution.

2. Background & Business Objective

- **Background of the Problem:** Traditional SME logistics follows a linear, manual workflow. An order is received, manual entry into a spreadsheet by the logistic executive, then the logistic executive decides on the number of drivers to dispatch. Route planning is usually based on the driver's local knowledge. External variables (storms, fuel spikes, road closures) are only addressed after they impact margins or service levels.

- **Importance of Solving This Issue:** Accelerating time required for a company to set up and manage logistics operations while letting them track their goods. Democratizing predictive logistics enables small businesses to compete with larger players, improve retention, and scale without hiring dedicated dispatch teams.

- **Strategic Fit / Impact:** Aligns with AI-Native operational workflows by combining open-source routing infrastructure, real-time public APIs, and lightweight LLMs. This demonstrates practical agentic AI design for supply chain resilience using free-tier services.

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3. Product Purpose

3.1. Main Goal of the System:

To provide an AI-enabled logistics co-pilot that evaluates pre-computed routes against real-world constraints and generates actionable, explainable recommendations for SME dispatchers.

3.2. Intended Users (Target Audience):

- SME Fleet Managers / Dispatch Coordinators
- Small Business Owner
- Entrepreneurs
- Last-Mile Operations Staff

4. System Functionalities

4.1. Description: The system operates as a hybrid decision engine. A routing solver (OpenRouteService) calculates baseline paths with distances, ETAs, and coordinates. The LLM agent ingests these paths alongside enriched external context (weather severity, fuel trends, disruption news), evaluates time-window and capacity constraints, and returns optimized sequences, risk mitigations, cost deltas, and customer messages. All LLM outputs are schema-validated before rendering.

4.2. Key Functionalities:

- **Dynamic Route Re-evaluation:** Context-aware sequence adjustment based on live weather, traffic, and fuel data without recalculating raw distances.
- **Risk Intelligence Dashboard:** Real-time alerting with severity levels, affected deliveries, and one-click mitigation steps.
- **Real-Time Vehicle and Goods Tracking:** Real-time tracking of shipments and driver allowing immediate contingency recommendations if unexpected accident occurs

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z AI GLM 5.1 is selected for high structured-output compliance, and cost-efficiency. The model excels at JSON schema enforcement and constraint reasoning, making them ideal for operational decision pipelines where hallucination tolerance is near zero.

4.3.2. **Prompting Strategy:** Our team uses a few-shot prompting strategy for higher accuracy, better formatting and more consistent results. We provide examples of intended output to get the AI to generate the desired JSON output.

4.3.3. **Context & Input Handling**: External API responses are pre-aggregated into concise JSON payloads before LLM injection. The maximum input size of our model should be 30 logistics hubs and 100 news articles before it needs to truncate the input. If the inputs are too big to be processed, it is truncated before accepting it.

4.3.4. **Fallback & Failure Behavior:** Our system will validate if the AI produced output in the correct format, and if not, the system will use fallback function to generate an output without LLM analysis. This can ensure basic reports will still be shown on the dashboard even during AI services' downtime.

5. User Stories & Use Cases

- **User Stories:** As a SME owner, I want to be able to track and manage logistics of my company without huge investment for a specific department. I want a system that can track shipments, research about potential risk and help me make business decisions for my company.
- **Use Cases (Main Interactions):**

  Dynamic Shipment Monitoring: Users can oversee various operations of the company such as warehouse package volume and capacity, fuel price, weather, news and other data.

  Dynamic Suggestion Generation: Users can generate suggestions that users can take to mitigate risk detected from various data points and AI analysis.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- Real-time integration with Open-Meteo, Malaysia Government Fuel Data, and MediaStackAPI.
- Structured LLM output (JSON) for sequence optimization, risk alerts, cost impact, and customer messages.
- Dashboard for route evaluation, operation report & cost analysis

7. Features Not Included (Scope Control)

- Implementation of login system and database
- Real-time GPS driver tracking, telematics, or live proof-of-delivery.
- Multi-user inputs simultaneously are not supported
- Deterministic routing engine (OSRM/Mapbox) for baseline path calculation.

8. Assumptions & Constraints

- **LLM Cost Constraint:** ~1.5K input + 500 output tokens per evaluation. Cost kept low by using flash models, caching static route segments, and pre-aggregating external APIs. Design decision: Context compression layer strips redundant API fields before LLM injection to minimize token burn.
- **Technical Constraints:** Routing relies on open-source OSRM or free-tier Mapbox. Frontend limited to React + Leaflet/Mapbox. State managed in-memory + lightweight JSON storage for hackathon demo.
- **Performance Constraints:** Max 20 deliveries per batch for optimal LLM evaluation latency (<3s). Generation limit: 3 replan cycles per session to prevent API throttling.
- **User Input:** System requires initial delivery data (addresses, priorities, time windows). AI does not auto-generate routes from scratch; it evaluates and optimizes pre-computed paths. Human approval required before applying any replan.

9. Risks & Questions Throughout Development

- **Data Latency:** Free-tier weather/news APIs may have delayed updates. Mitigation: Use cached fallback values + pre-loaded mock datasets for demo reliability.
- **LLM Constraint Violation:** Risk of LLM suggesting impossible swaps (e.g., exceeding capacity or breaking time windows). Mitigation: Strict Pydantic validation + post-generation constraint checker before rendering.
- **Routing Engine Fallback:** If OSRM returns no route due to geocoding errors, system defaults to straight-line distance approximation with a UI warning banner.
- **Question:** How do we handle time-zone mismatches for multi-region SME deliveries? (Scope limited to single-region/day for v1; multi-zone queued for post-hackathon)
- **Question:** What happens if fuel price API returns stale or missing data? (System uses last-known trend with confidence scoring in LLM output; explicitly labeled as "estimated" in UI.)